

$$1) f(x) = 5 + 2x^2$$

$$W = \int_0^4 f(x) dx$$

$$a = 0$$

$$b = 4$$

$$n = 4$$

$$\Delta x = \frac{b-a}{n}$$

$$= \frac{4-0}{4} = 1$$

$$x_i^* = a + \left(i - \frac{1}{2}\right) \Delta x$$

$$x_1^* = 0,5$$

$$f(0,5) = 5 + 2(0,5)^2$$

$$= 5,5$$

$$x_2^* = 1,5$$

$$f(1,5) = 5 + 2(1,5)^2$$

$$= 9,5$$

$$x_3^* = 2,5$$

$$f(2,5) = 5 + 2(2,5)^2$$

$$= 17,5$$

$$x_4^* = 3,5$$

$$f(3,5) = 5 + 2(3,5)^2$$

$$= 29,5$$

$$W = \Delta x [f(x_1^*) + f(x_2^*) + f(x_3^*) + f(x_4^*)]$$

$$W = 1 (5,5 + 9,5 + 17,5 + 29,5)$$

$$= 62$$

$$2) C(T) = 20 + 0,05T^2$$

$$Q = \int_0^{100} C(T) dT$$

$$a = 0 \quad b = 100$$

$$b = 100$$

$$\Delta T = \frac{100-0}{4} = 25$$

$$T_0 = 0$$

$$C(0) = 20$$

$$T_1 = 25$$

$$C(25) = 20 + 0,05(25)^2$$

$$= 51,25$$

$$T_2 = 50$$

$$C(50) = 20 + 0,05(50)^2$$

$$= 145$$

$$T_3 = 75$$

$$C(75) = 20 + 0,05(75)^2$$

$$= 301,25$$

$$T_4 = 100$$

$$C(100) = 20 + 0,05(100)^2$$

$$= 520$$

$$Q \approx \frac{\Delta T}{2} [C(T_0) + 2C(T_1) + 2C(T_2) + 2C(T_3) + C(T_4)]$$

$$= \frac{25}{2} [20 + 2(51,25) + 2(145) + 2(301,25) + 520]$$

$$= \frac{25}{2} (1935) = 19187,5$$

$$3) I(t) = 2 + 3t + t^2$$

$$a \rightarrow \int_0^6 I(t) dt$$

$$a=0$$

$$b=6$$

$$n=6$$

$$\Delta T = \frac{6-0}{6} = 1$$

$$t_0=0$$

$$I(0)=2$$

$$t_1=1$$

$$I(1) = 2 + 3(1) + (1)^2 \\ = 6$$

$$t_2=2$$

$$I(2) = 2 + 3(2) + (2)^2 \\ = 12$$

$$t_3=3$$

$$I(3) = 2 + 3(3) + (3)^2 \\ = 20$$

$$t_4=4$$

$$I(4) = 2 + 3(4) + (4)^2 \\ = 30$$

$$t_5=5$$

$$I(5) = 2 + 3(5) + (5)^2 \\ = 42$$

$$t_6=6$$

$$I(6) = 2 + 3(6) + (6)^2 \\ = 56$$

$$a = \frac{\Delta t}{3} [I(t_0) + 4I(t_1) + 2I(t_2) + 4I(t_3) + 2I(t_4) + 4I(t_5) + I(t_6)]$$

$$= \frac{1}{3} [2 + 4(6) + 2(12) + 4(20) + 2(30) + 4(42) + 56]$$

$$= \frac{1}{3} [2 + 24 + 24 + 80 + 60 + 168 + 56]$$

$$= \frac{1}{3} (414) = 138$$